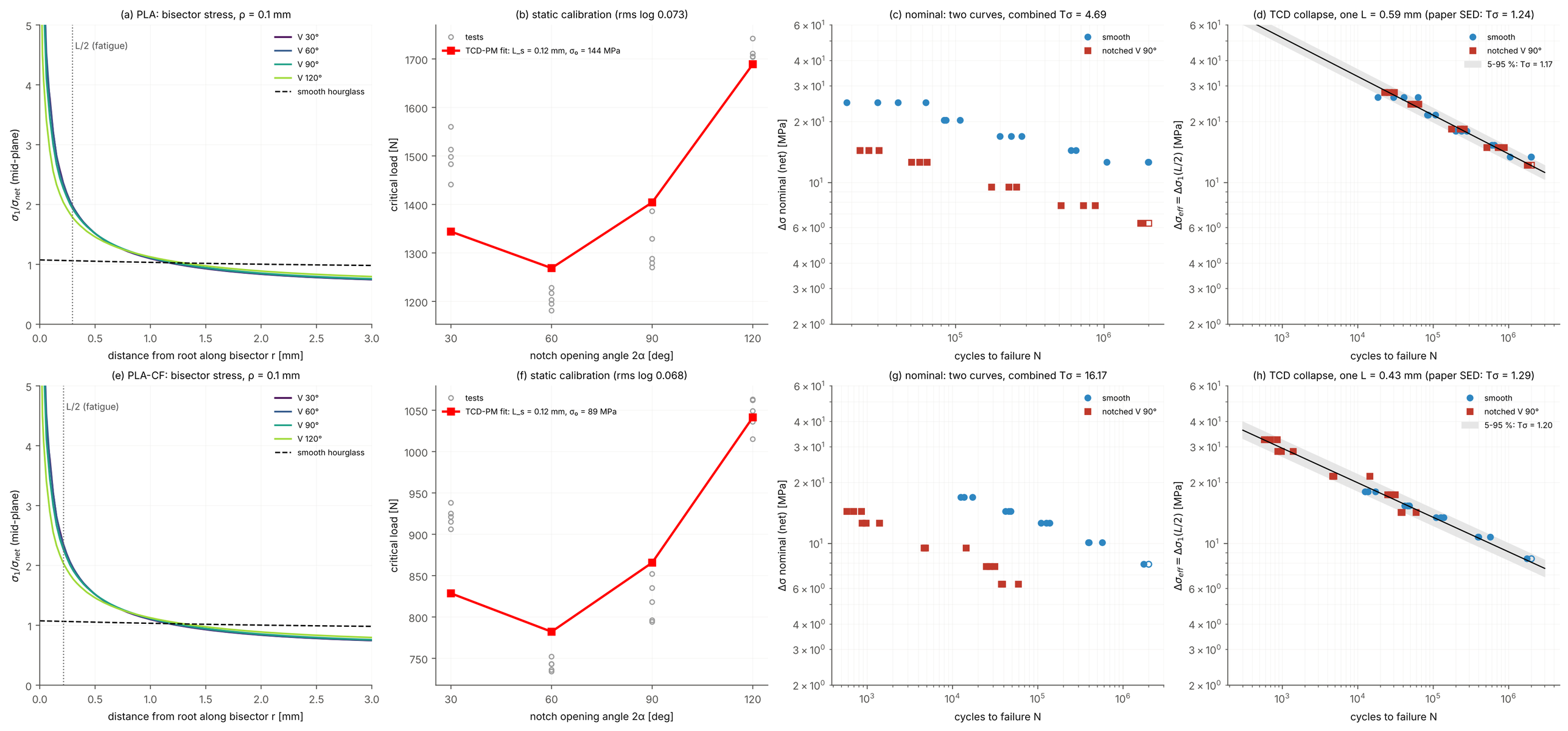

(a) PLA: bisector stress, ρ = 0.1 mm
L/2 (fatigue)
V 30°
V 60°
V 90°
V 120°
smooth hourglass
σ1/σnet (mid-plane)
distance from root along bisector r [mm]
(b) static calibration (rms log 0.073)
tests
TCD-PM fit: L_s = 0.12 mm, σo = 144 MPa
critical load [N]
notch opening angle 2α [deg]
(c) nominal: two curves, combined Tσ = 4.69
smooth
notched V 90°
Δσ nominal (net) [MPa]
cycles to failure N
(d) TCD collapse, one L = 0.59 mm (paper SED: Tσ = 1.24)
smooth
notched V 90°
5-95 %: Tσ = 1.17
Δσeff = Δσ1(L/2) [MPa]
cycles to failure N
(e) PLA-CF: bisector stress, ρ = 0.1 mm
L/2 (fatigue)
V 30°
V 60°
V 90°
V 120°
smooth hourglass
σ1/σnet (mid-plane)
distance from root along bisector r [mm]
(f) static calibration (rms log 0.068)
tests
TCD-PM fit: L_s = 0.12 mm, σo = 89 MPa
critical load [N]
notch opening angle 2α [deg]
(g) nominal: two curves, combined Tσ = 16.17
smooth
notched V 90°
Δσ nominal (net) [MPa]
cycles to failure N
(h) TCD collapse, one L = 0.43 mm (paper SED: Tσ = 1.29)
smooth
notched V 90°
5-95 %: Tσ = 1.20
Δσeff = Δσ1(L/2) [MPa]
cycles to failure N